Individual Contribution Report

Student Name: Neil Patrick Sermeno
Group: Shared Brain Cells
Assigned Role: Process and Memory Developer

What did I contribute to the project?

I was in charge of the process and memory part of Shared Brain Cell. I wrote processes.py, which keeps a table of running processes with PIDs, and memory.py, which tracks total, used, and free memory. I also made the commands run, process, kill, memory, allocate, and free, and worked with Kian so they plugged into main.py

What features did I personally develop?

Processes: run starts a fake process with its own PID, process lists them, and kill ends one by PID

Memory: memory shows total, used, and free. allocate uses some MB and free gives it back

Error handling: bad PIDs and bad memory amounts show an error message instead of crashing

What problems did I encounter?

Keeping PIDs unique after a process was killed

Users typing wrong input, like a PID that doesn't exist or letters instead of a number

Keeping the memory numbers correct when allocating or freeing too much

Making my code work with the other modules

How did I solve those problems?

I used a counter that only goes up and stored processes by PID, so no two share a number.

I check every input first and print a clear error if it's wrong.

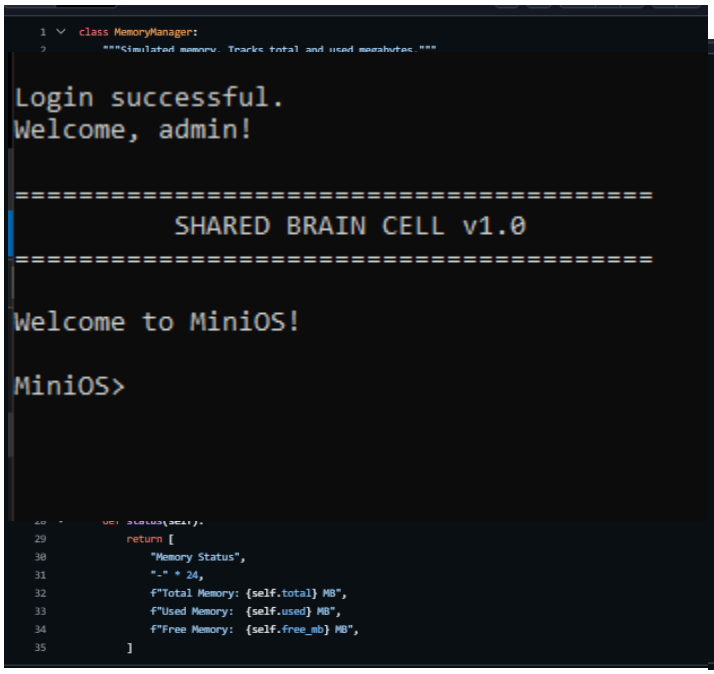
allocate and free refuse amounts that are too big, and free memory is always total minus used.

I made my functions return simple results for main.py to print, and tested with my groupmates.

What did I learn from this project?

I learned how an operating system tracks processes and memory, and why PIDs are needed. I also learned to validate input so one bad command doesn't crash the program, and how to combine everyone's code with GitHub.

Evidence of my contribution:



```
class MemoryManager:
    """Simulated memory. Tracks total and used megabytes."""

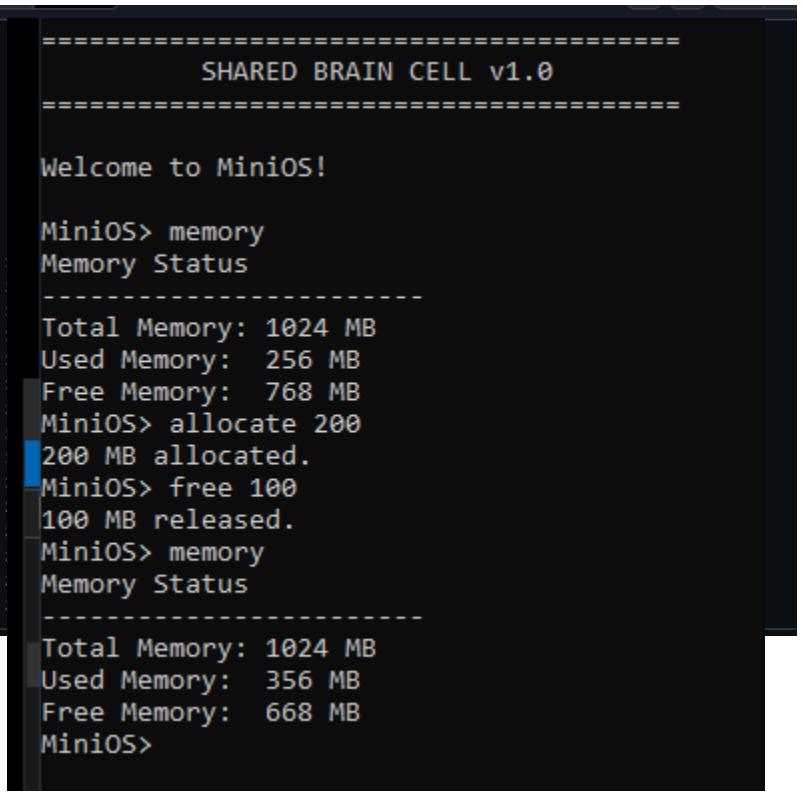
Login successful.
Welcome, admin!

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          SHARED BRAIN CELL v1.0
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Welcome to MiniOS!

MiniOS>
```

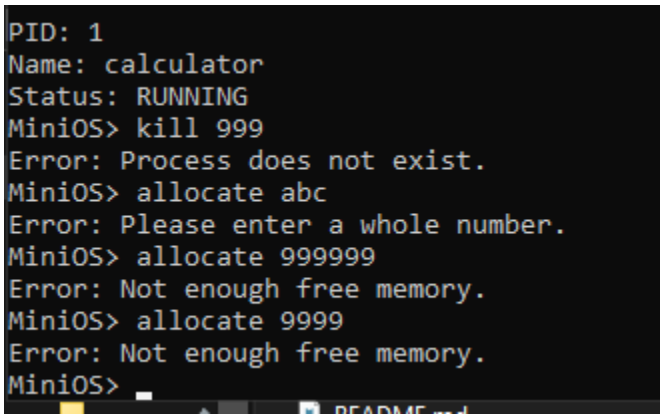
```
        return [
            "Memory Status",
            "-" * 24,
            f"Total Memory: {self.total} MB",
            f"Used Memory:  {self.used} MB",
            f"Free Memory:  {self.free_mb} MB",
        ]
```



```
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          SHARED BRAIN CELL v1.0
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Welcome to MiniOS!

MiniOS> memory
Memory Status
------------------------
Total Memory: 1024 MB
Used Memory:  256 MB
Free Memory:  768 MB
MiniOS> allocate 200
200 MB allocated.
MiniOS> free 100
100 MB released.
MiniOS> memory
Memory Status
------------------------
Total Memory: 1024 MB
Used Memory:  356 MB
Free Memory:  668 MB
MiniOS>
```



```
PID: 1
Name: calculator
Status: RUNNING
MiniOS> kill 999
Error: Process does not exist.
MiniOS> allocate abc
Error: Please enter a whole number.
MiniOS> allocate 999999
Error: Not enough free memory.
MiniOS> allocate 9999
Error: Not enough free memory.
MiniOS>
```